Miguel Angel Velásquez Vásquez

Personal Information:

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Education

31/12/2024 PhD in Psychology, Tulane University.
Dissertation: Investigating the effects of musical training and neurofeedback on cognitive enhancement and brain plasticity. *Advisor:* Dr. Paul J. Colombo.

07/05/2021 M.Sc. in Psychology, Tulane University.
Thesis: The effects of neural entrainment on sensorimotor synchronization. *Advisor:* Dr. Paul J. Colombo.

09/05/2014 B.S. in Psychology with Honors, University of New Orleans.
Thesis: Reciprocal relation between psychophysiological patterns of stress responsivity and sleep. *Advisor:* Dr. Elizabeth A. Shirtcliff .

Research Experience

2025 - Ongoing **Postdoctoral Researcher** Department of Psychology, University of New Orleans.
Supervisor: Dr. Debra Karhson

- Assisted with the data analysis and dissemination of research results including creation of grants, manuscripts, and presentations.
- Collected EEG, saliva and blood analytes, and behavioral data from autistic and neurotypical participants.
- Overseeing scheduling and tracking of participants, quality control of local data collection, & data management and data transfer.
- Developed efficient data cleaning and analysis pipelines of periodic and aperiodic EEG data components.

2024 **Researcher Assistant** Department of Economy, Tulane University.
Supervisor: Dr. Patrick Button & Dr. Brigham Walker

- Conducted literature reviews and statistical analyses on factors contributing to race-related accessibility issues in mental health care.

2018-2024 **Researcher Assistant** Department of Psychology, Tulane University.
Supervisor: Dr. Paul Colombo

- Conducted data analysis and dissemination of research results including management of grant-funded projects, writing manuscripts, and presentations.
- Conducted a meta-analysis on the relationship between neurofeedback training parameters and neural modulation success.
- Performed ERP and behavioral analyses to determine the differences in inhibitory control between musicians and non-musicians.
- Collaborated on the experimental design, collection and analysis of fNIRS data to determine the impact of acute and chronic mindful meditation on hemodynamic brain activity.
- Collected and analyzed EEG and behavioral data on 60 participants to determine the prolonged effects of auditory stimuli on SSEP signals.
- Conducted cortisol and total protein extraction on saliva samples using ELISA method to test the effects of music training on stress reactivity.

2014 **Researcher Assistant** Department of Psychology, University of New Orleans.
Supervisor: Dr. Connie Lamm

- Operated as a Dense Array EEG Technician.
- Collected neurophysiological, and behavioral data from human participants.
- Performed data cleaning and entry on collected data.

2011 - 2014 **Researcher Assistant** Department of Psychology, University of New Orleans.
Supervisor: Dr. Elizabeth Shirtcliff

- Studied the effects of sleep quality on stress reactivity and salivary cortisol.
- Collected biological and behavioral data from human participants.
- Wet lab technician performing enzyme-immuno assays (ELISA) on salivary samples.

Publications

(*Denotes equal contribution to the work)

2025 Galang, E. J.*, **Velasquez, M. A.***, Elcin, D., O'Connell, S., & Colombo, P. J. (2025). Systematic Review and Meta-Analysis of the Relationships between Real-Time Neurofeedback Training Parameters and Acquisition of Neural Modulation. *Frontiers in Human Neuroscience, 19*, 1652607.

2024

Velasquez, M. A., Winston, J. L., Sur, S., Yurgil, K. A., Upman, A. E., Wroblewski, S. R., Huddle, A., & Colombo, P. J. (2024). Music training is related to late ERP modulation and enhanced performance during the Simon task but not the Stroop task. *Frontiers in Psychology, 18*, 1384179.

2024 Elcin, D., **Velasquez, M. A.**, & Colombo, P. J. (2024). Effects of acute and long-term mindfulness on the conflict resolution component of attention. *Frontiers in Human Neuroscience, 18*, 1359198.

2020 Yurgil, K. A., **Velasquez, M. A.**, Winston, J. L., Reichman, N. B., & Colombo, P. J. (2020). Music training, working memory, and neural oscillations: A review. *Frontiers in Psychology, 11*, 266.

Works in Progress

2026 **Velasquez, M. A.**, Weinstein, C. R., Scott, M. V., & Karhson, D. (In Preparation). Auditory task-related aperiodic EEG activity in autism spectrum disorder (ASD).

2026 Brigham, W., **Velasquez, M. A.**, & Button, P. (In Preparation). Role of race concordance in mental health access.

Participation at Professional Meetings

(*Denotes collaborator-led presentations)

2026 "Auditory task-related aperiodic EEG activity in autism spectrum disorder (ASD)." *Poster presented at the Cognitive Neuroscience Society Conference*. Vancouver, Canada

2023 "Music training is related to late ERP modulation and enhanced performance during the Simon task but not the Stroop task." *Poster presented at Society for Neuroscience Conference*. Washington, D.C.

2023 *"Effects of acute and long-term mindfulness on the conflict resolution component of attention." *Poster presented at Society for Neuroscience Conference*. Washington, D.C.

2023 "The influence of musicianship on the timing of cognitive conflict resolution in Stroop and Simon tasks." *Poster presented at Tulane Research, Innovation and Creativity Summit*. New Orleans, LA.

2023 *"Effects of acute and long-term mindfulness on the conflict resolution component of attention." *Poster presented at Tulane Research, Innovation and Creativity Summit*. New Orleans, LA.

2022 "Rhythmic priming results in neural entrainment and persistent effects on sensorimotor synchronization." *Poster presented at Health Science Research Days, Tulane University*. New Orleans, LA.

2021 "Rhythmic priming results in neural entrainment and persistent effects on sensorimotor synchronization." *Poster presented at Society for Neuroscience Conference*. Chicago, IL.

2021 "Musical experience is related to the time course of changes in levels of the immune response marker interleukin-6 after exposure to acute social stress." *Poster presented at Neuromusic VII – Fondazione Mariani*. Aarhus, Denmark.

2015 *"A neurobiological view on testosterone responsivity to reward and challenge." *Presented at the International Convention of Psychological Science (ICPS).* Amsterdam, The Netherlands.

2014 "Reciprocal relation between psychophysiological patterns of stress responsivity and sleep." *Presented at Posters on the Hill*. Washington, D.C.

2014 "Reciprocal relation between psychophysiological patterns of stress responsivity and sleep." *Poster presented at Psychology Undergraduate Research Conference*. UCLA, Los Angeles, CA.

Invited Talks

2026 "The effects of musical training and neurofeedback on cognitive enhancement and brain plasticity." Psychology Graduate Seminar Series, University of New Orleans .

2026 "Auditory task-related aperiodic EEG activity in autism spectrum disorder (ASD)" Cognitive Neuroscience Society Data Blitz, Vancouver.

2022 "The effects of neural entrainment on sensorimotor synchronization." The Robert E. Flowerree III Psychology Colloquium Series, Tulane University.

Honors & Awards

2026 Marie Skłodowska-Curie Actions (MSCA) Seal of Excellence (94.4%).

2026 Post-Doctoral Fellow Award. Cognitive Neuroscience Society 2026

2014 Excellence in Psychology Award. 41st annual College of Sciences Honors Convocation

2013 S. Thomas Elder Award Most Promising Undergraduate Research. Annual College of Sciences Honors Convocation

2012 Supervised Undergraduate Research Experience (SURE). Experimental program to Stimulate Competitive Research (LAEPSCoR).

Teaching: Mentorship

Graduate Student Supervision
Charlie Weinstein, Mya Scott

Honors Thesis Student Supervision
Charlotte Pearson

Baccalaureate Student Supervision
Laura Domensch, Eldrick Galang, Conner Angelle, Ilyssa Muise, Abby Freedman, Savannah McNair, Liam Tierney

AP Research Mentor
Eesha Dasai

Skills

Programming and Statistical Tools: Matlab, R, Python, SPSS

Neuroimaging Tools: EEG, fNIRS, ERP
Signal Processing: Time-frequency analysis, power spectral density (PSD), dynamic and static functional connectivity, steady state evoked potentials (SSEP)
Wet Lab Techniques: Enzyme-linked immunosorbent assay (ELISA)
Others: E-Prime, PsychoPy, BCI

Languages: Spanish (Native), English (Fluent), French (Conversational)

Service to the Field

Ad Hoc Journal Reviewer
Frontiers in Psychology - Auditory Cognitive Neuroscience
Annals of the New York Academy of Sciences
Nature - Scientific Reports
Behavioral and Brain Functions Advances in Cognitive Psychology

Membership in Professional Societies

Cognitive Neuroscience Society (CNS)
Society for Neuroscience (SfN)
America Psychological Association (APA)
Society for Music Perception and Cognition (SMPC)
Multidisciplinary Association for Psychedelic Studies (MAPS)

Other

2023 **Siddhartha Opera** Interactive Music and Dance Performance, Music Co-composer

2023 **Walk-way Voices** Interactive Sound Art Installation, Greenway Supernova

2023 **First LEGO League Challenge Championship Judge**

2022 **Finalist in Novel Tech Challenge technology development program** "Neurowaves: gamma wave neural stimulation through audiovisual stimulation"